INF1001 – Introduction to Computing

PART 1 LAB

Objectives

1. Demonstrate the effect of sampling and quantization.
2. Verify Digital circuits using simulation.
3. Design and verify Digital circuits for a real-world problem.

Assessment – Lab Report Submission

The assessment will be based on the Lab Report. Each student has to submit a Lab Report in the specific folder (Folder name is "INF1001 Lab Report (Part 1) Submission Folder") created on xsite latest by the end of Week 7, i.e., **23:59 hours on 22 February 2026**. The details to be presented in the Lab Report are mentioned in the respective sections of this document. The first page (cover page) of the lab report must indicate your name, roll number, module code, lab name (i.e., Lab-1) and submission month (i.e., February/March 2024). The Lab Report should be in PDF format (No other files are acceptable).

Plagiarism

The University's policy on copying does not allow you to copy software as well as your assessment solutions from another person. It is the responsibility of all students that their assessment solutions are their own work. If plagiarism is detected, the assessment involved will receive ZERO marks.

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1. Sampling and Quantization

Sampling is the process of recording an analog signal at regular discrete moments of time. The sampling rate (frequency) *fs* is the number of samples per second. The time interval between samples is called the sampling interval $Ts = 1/fs$.

The **Nyquist-Shannon sampling theorem** states that the sampling rate for exact recovery of a signal composed of a sum of sinusoids is larger than twice the maximum frequency of the signal.

Quantization:

In order to complete analog to digital conversion, each sample value is mapped to a discrete level (represented by a sequence of bits) in a process called **quantization.** In an *n*-bit quantizer, each quantization level is represented with *n* bits, so that the number of levels equals 2^n.

Quantization step-size, $Q = \frac{FS}{2^n}$, where *FS* is the full-scale span of ADC and *n* is the number of bits.

Maximum Quantization error = Q/2.

Exercise 1.1

In this online exercise, you will listen and 'feel' the effect of sampling and quantization using audio signals. The exercise demonstrates the effect of sampling audio signal using different sampling frequencies. The online demonstration shows the effect of quantization of an audio signal sampled at 44,100 Hz using 16 bits, 8 bits, 6 bits, 4 bits, 3 bits and 2 bits. You can play the audio signal corresponding to each of these quantization levels (number of bits) and feel the effect. The other part of this exercise will demonstrate the effect of using different sampling frequencies (44,100 Hz, 22,500 Hz, 11,025 Hz, 4410 Hz, 2205 Hz, 1407 Hz, 882 and 441 Hz).

You can access the online exercise here:
https://makeabilitylab.github.io/physcomp/signals/QuantizationAndSampling/index.html

(a) Sampling: On the above website, follow Sections 4.4, 4.4.1, 4.4.2, 4.4.3, 4.4.4, 4.4.5, 4.4.6 and 4.4.7.
Comment on the effect of sampling the audio signal using different sampling frequencies. Briefly explain your observation and summarize the findings.

(b) Quantization: Follow Sections 3.2, 3.2.1, 3.2.2, 3.2.3, 3.2.4 and 3.2.5

Comment on the effect of quantizing the audio signal using different word lengths (number of bits). Briefly explain your observation and summarize the findings.

Lab Report: Include your comments on the effect of sampling and quantizing the audio signal using different sampling rates and word lengths respectively.

2. Verify Digital Circuits using Simulation

For this part of the lab exercise, you will use the free online digital circuit simulator **'circuitVerse'**.

https://circuitverse.org/simulator

CircuitVerse is an easy-to-use digital logic circuit simulator that aims at providing a platform to create, share and learn digital circuits. It can run on almost any device without the need for installing any software (cloud-based open-source tool).

Watch this **TUTORIAL VIDEO** on CircuitVerse simulator features and an example of how to build and simulate a logic circuit:
https://www.youtube.com/watch?v=F-_uvYJK938

Exercise 2.1

Building and verification of logic circuit

1. Launch circuitverse.org/simulator and build the logic circuit for realizing the Boolean expression:

$$X = ABC + \overline{A}BC + AC + A\overline{C}$$

2. Construct the TRUTH TABLE for 'X'.

3. Implement the Boolean expressions in CircuitVerse and Verify the Truth Table by simulating respective input values of A, B and C and checking the output X of the simulated circuit.

4. Simplify the Boolean expression X so as to implement it using only ONE AND gate and ONE OR gate. Construct the TRUTH TABLE of the simplified Boolean expression for X.

5. Verify the Truth Table of the simplified X by simulating respective input values of A, B and C and checking the output X of the simulated circuit.

Lab Report: Your lab report should include (a) Truth Table of the original Boolean expression X, (b) Screenshots of the CircuitVerse simulator showing the verification of the implemented circuit with respect to the Truth Table, (c) Simplified Boolean expression and its Truth Table, and (d) Screenshots of the CircuitVerse simulator showing the verification of the implemented simplified circuit with respect to its Truth Table. All the input combinations need to be tested and verified.

3. Design and verify Digital circuits for a Real-world Problem

Line Follower Robot

A Line Follower Robot (LFR) is a simple autonomously guided robot that follows a line drawn on the ground to detect a dark (black) line on a white surface. You may refer to the Lecture Slides for the details of the principle of operation of the LFR and its control logic. Mainly, the LFR control logic can be summarized as the FOUR SCENARIOS described below.

Scenario 1: When the robot moves in the correct path along the center, the IR sensors are aligned at the top of the white surface. RX of 'L' and 'R' sensors receive reflected light and the transistors T_L and T_R connected to them turns ON → Outputs of both transistors become LOW ($T_L = 0$, $T_R = 0$). This indicates that the robot is in the correct track and the controller will turn ON LEFT and RIGHT MOTORS of the robot $M_L=1$ and $M_R=1$ → Robot moves forward.

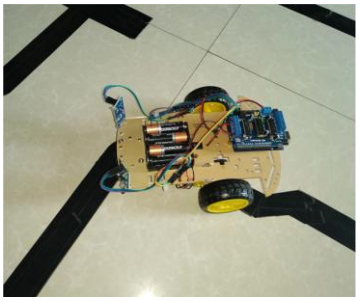

Fig. 1. Line following robot (LFR).

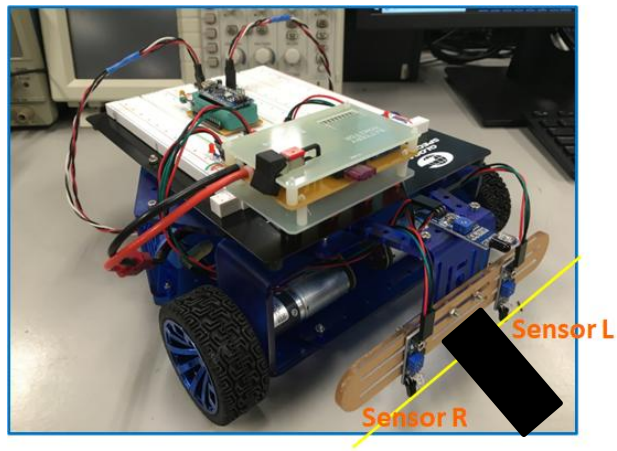

Fig. 2. Sensors of LFR. Black line at the center is the track and the Left (L) & Right (R) sensors detect white/black surface.

Scenario-2: When the robot shifts to the left, the 'L' sensor is still on white surface but the 'R' sensor is on top of black surface. RX of 'L' receives light whereas RX of 'R' does not receive light. T_L connected to 'L' sensor is LOW (0) and T_R connected to 'R' sensor is HIGH (1) → (T_L = 0, T_R = 1). This indicates that the robot is shifting towards left of the track → controller will turn OFF RIGHT MOTOR and turn ON LEFT MOTOR M_L=1 and M_R =0 → Robot moves towards RIGHT back to the center-line.

Scenario-3: When the robot shifts to the right, the 'R' sensor is still on white surface but the 'L' sensor is on top of black surface. RX of 'R' receives light whereas RX of 'L' does not receive light. T_L connected to 'L' sensor is HIGH (1) and T_R connected to 'R' sensor is LOW (0) → (T_L = 1, T_R = 0). This indicates that the robot is shifting towards right of the track → controller will turn OFF LEFT MOTOR and turn ON RIGHT MOTOR M_L=0 and M_R =1 → Robot moves towards LEFT back to the center-line.

Scenario-4: If the robot makes a 90-deg turn to LEFT or RIGHT, both 'L' and 'R' sensors will be aligned on top of black surface. RX of 'L' and 'R' do not receive light. T_L connected to 'L' sensor is HIGH (1) and T_R connected to 'R' sensor is HIGH (1) → (T_L = 1, T_R = 1). This indicates that the robot has made a 90-deg turn away from the track → controller will turn OFF both RIGHT MOTOR and LEFT MOTOR M_L=0 and M_R =0 → Robot will STOP.

<u>Exercise 3.1</u>

1. Construct the TRUTH TABLE for the LEFT Motor and RIGHT Motor control signals (M_L and M_R respectively) in terms of respective inputs T_L and T_R.

2. Obtain the Boolean expressions for M_L and M_R.

3. Implement the Boolean expressions in CircuitVerse and verify the Truth Tables by simulating the respective input values of T_L, and T_R and checking the respective outputs M_L and M_R of the simulated circuits. You may connect LEDs (available in the 'circuit elements' tool box under the 'output' category) at the output points M_L and M_R of the logic circuit.

Lab Report: Your lab report should include (a) Truth Tables of the Boolean expressions for M_L and M_R and (b) Screenshots of the CircuitVerse simulator showing the verification of the implemented circuits with respect to the Truth Table. All the input combinations need to be tested and verified.

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